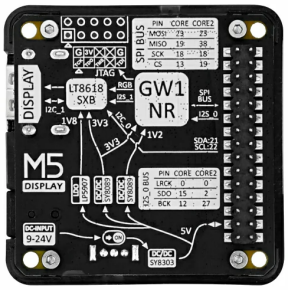
Module13.2 Display

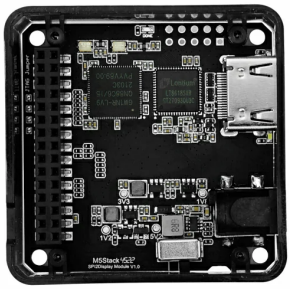
SKU:M126

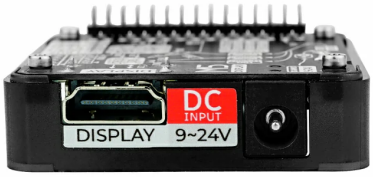


















Description

Module13.2 Display is an expansion module designed for **high-definition audio and video**. It uses the **Gowin GW1NR series FPGA chip** to output **Display signals** and the **LT8618S chip** for signal conditioning, enabling **24-bit color depth Display signals** and **up to 8-channel audio signals** with a resolution of up to **1080P**. The FPGA internally implements the **I2S audio generation function**, and the module includes a **DC power input socket** and corresponding **DC-DC circuits** for powering the entire device. This product is suitable for **various programmable high-definition audio and video output** applications.

Features

- Supports 24-bit RGB input
- Resolution up to 1080P
- Supports up to 8-channel audio input
- Includes 9-24V to 5V, 3.3V, and 1.2V DC-DC circuits and 5V to 1.8V LDO circuit
- High-definition audio and video interface, JTAG download interface
- 9-24V DC power interface
- Power switch

Includes

- 1 x Module13.2 Display

Applications

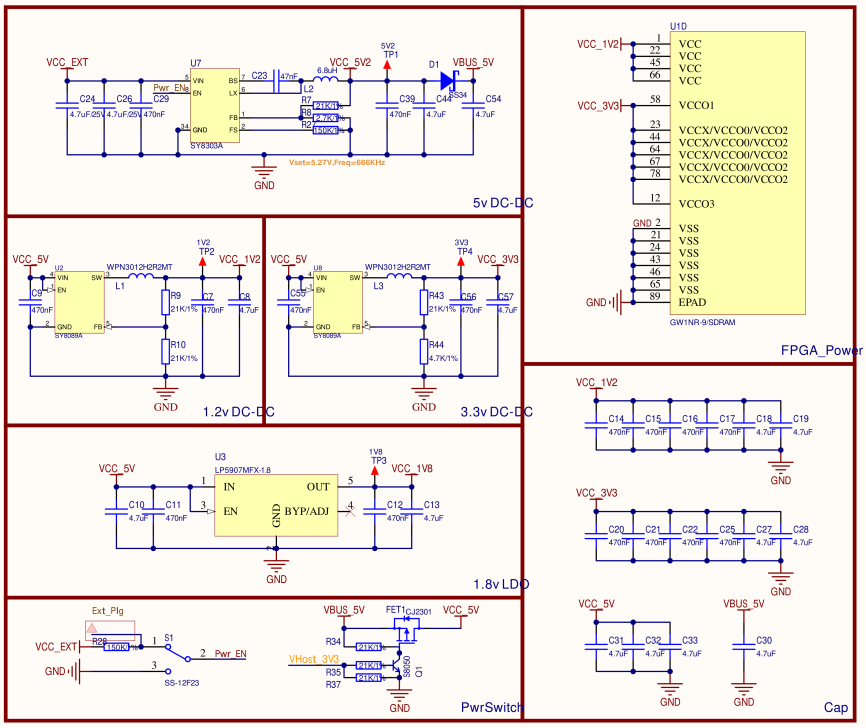
- Output high-definition audio and video signals
- Human-machine interaction

Specifications

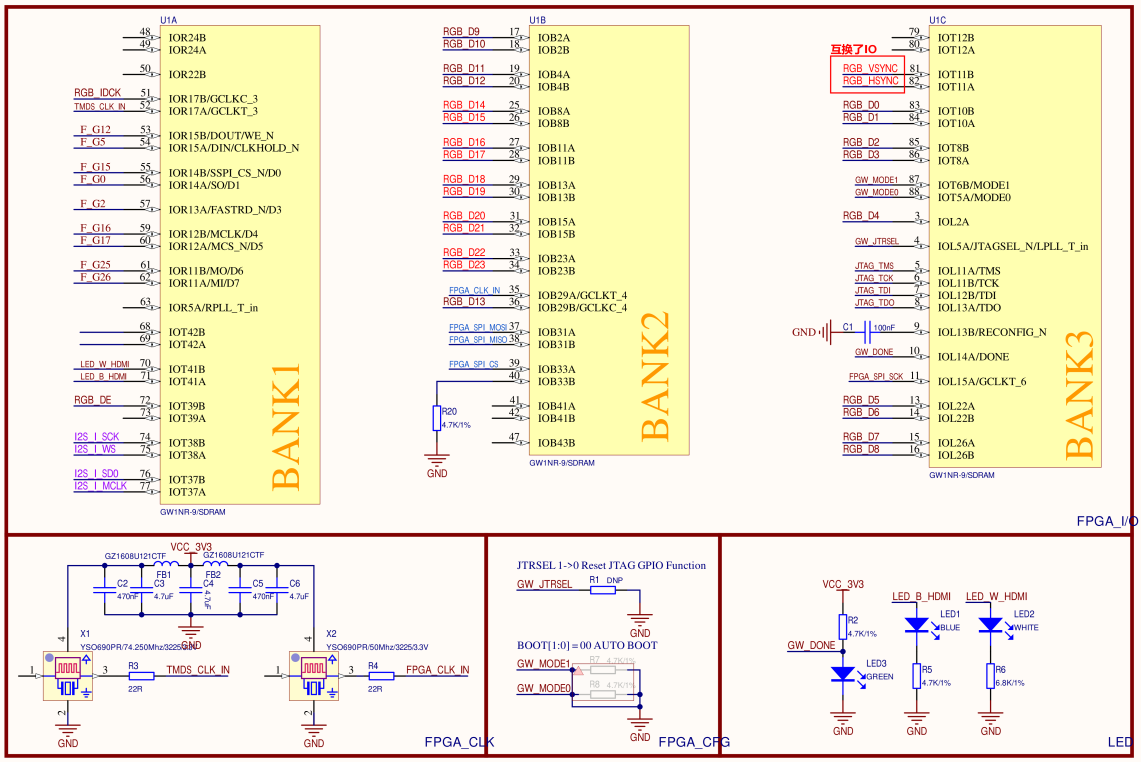
Specification	Parameter
Audio/Video Signal Acquisition Chip	GW1NR-LV9QN88C6-I5 FPGA
Audio/Video Signal Processing Chip	LT8616SXB @ 0x39
9-24V to 5V DC-DC Buck Chip	SY8303
5V to 3.3V and 1.2V DC-DC Buck Chip	SY8089
5V to 1.8V LDO Buck Chip	LP5907
Resolution	Up to 1080P (1920x1080)
Power Input	9 ~ 24V
Product Size	54.0 x 54.0 x 19.8mm
Product Weight	21.6g
Package Size	134.0 x 95.0 x 20.2mm
Gross Weight	41.2g

Schematics

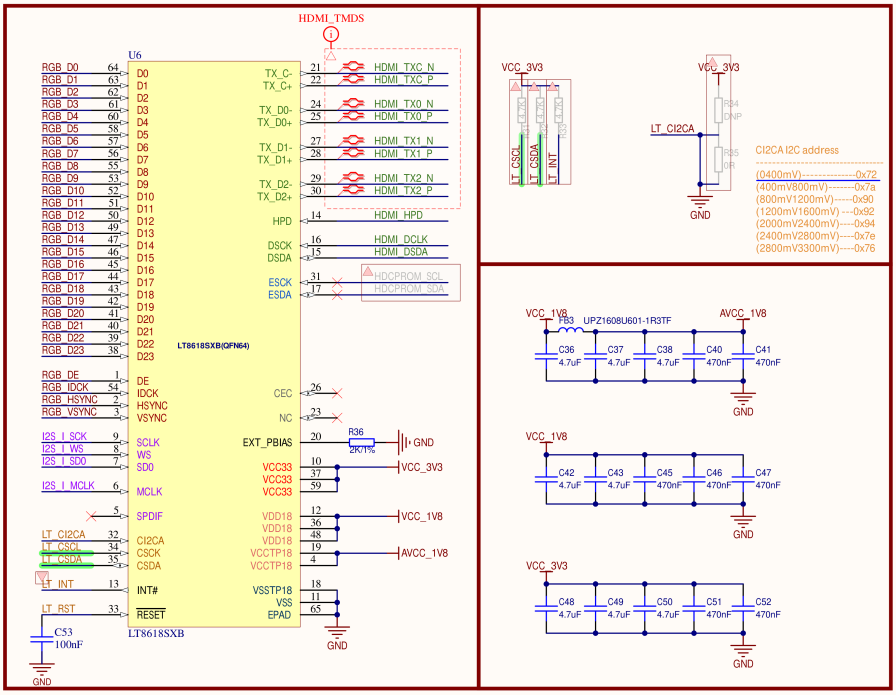
- Module13.2 Display Schematics PDF



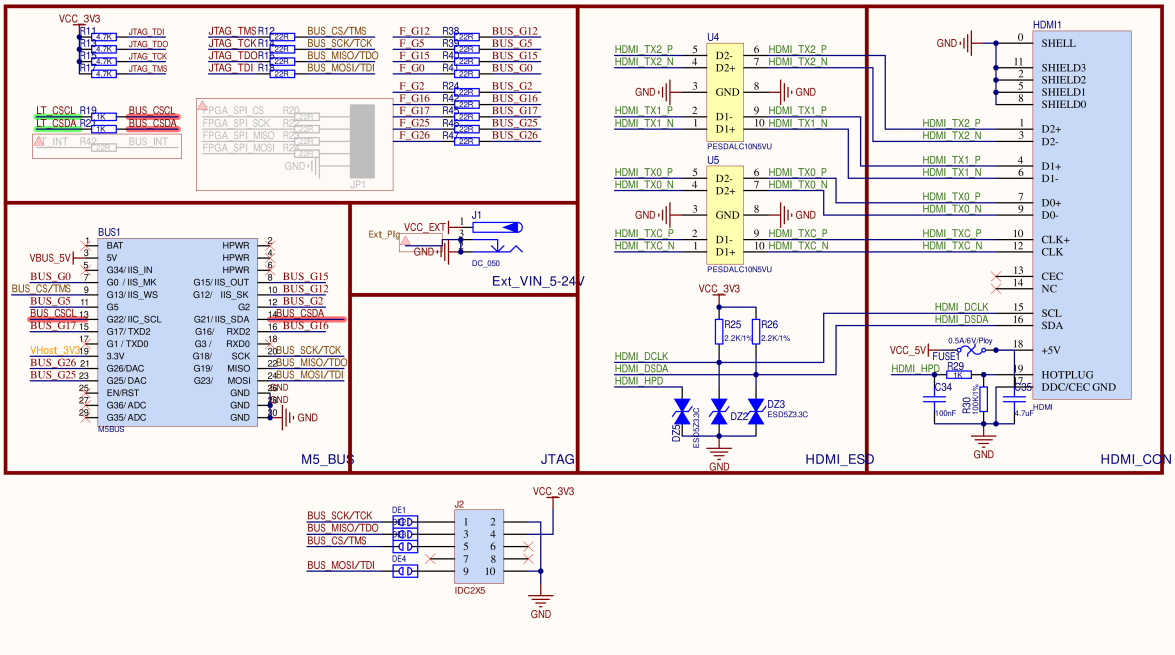
Title		
Size A4	Number	Revision
Date: 11/08/2022	Sheet of	
File: F:\AltiumDesigner\..\PSU.SchDoc	Drawn By:	



Title		
Size A4	Number	Revision
Date: 11/08/2022	Sheet of	
File: F:\AltiumDesigner\..\FPGA.SchDoc	Drawn By:	



Title		
Size A4	Number	Revision
Date: 11/08/2022	Sheet of	
File: F:\AltiumDesigner\..\RGB2HDMI.SchDoc	Drawn By:	



Title		
Size A4	Number	Revision V0.2
Date: 11/08/2022	Sheet of	
File: F:\AltiumDesigner\..\PINOUT.SchDoc	Drawn By:	

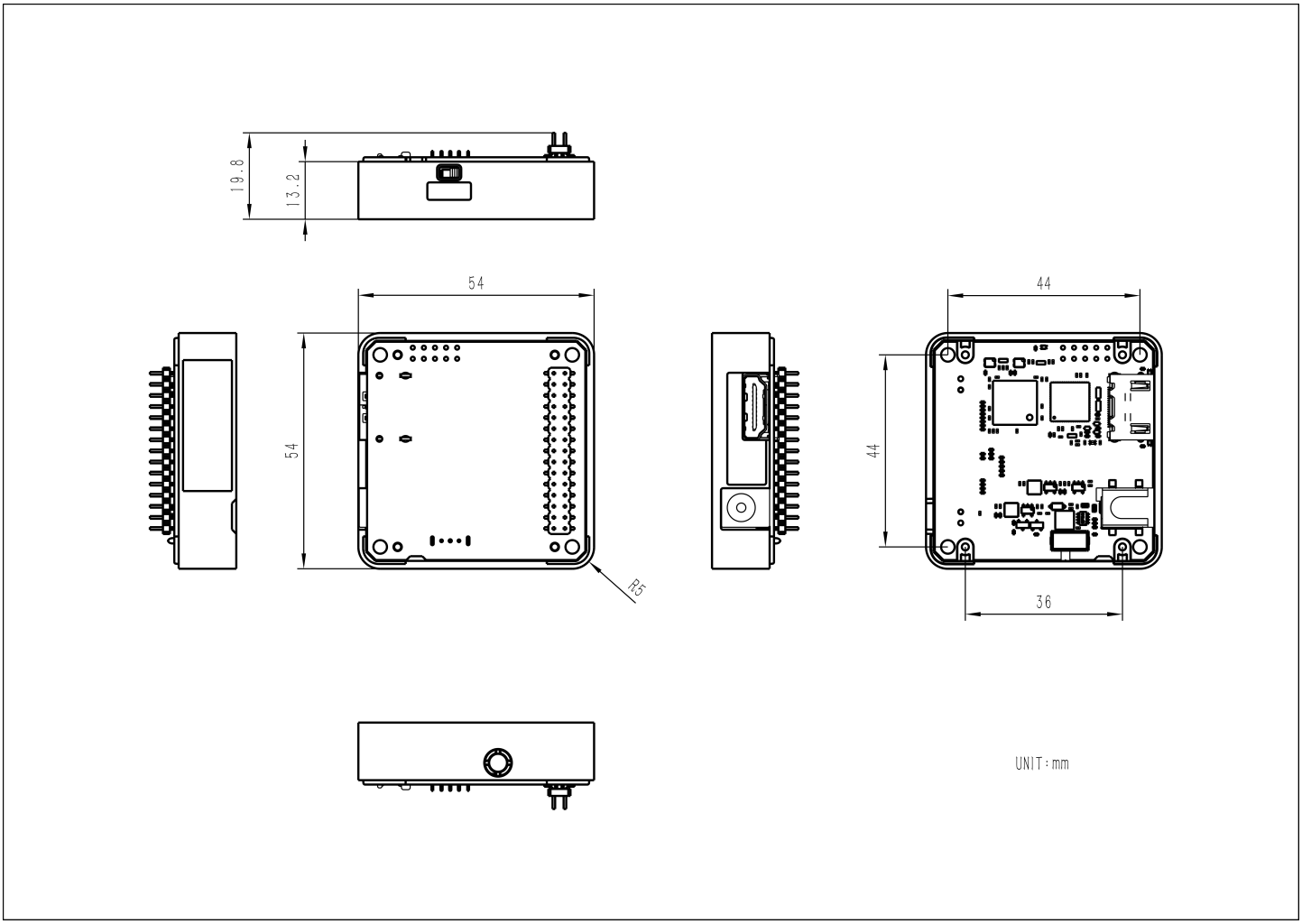
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
SPI_MOSI	7	8	
SPI_MISO	9	10	
SPI_CLK	11	12	3V3
	13	14	
	15	16	
I2C_SDA	17	18	I2C_SCL
	19	20	
	21	22	
	23	24	
	25	26	
	27	28	5V
	29	30	

Model Size

- Module13.2 Display Model Size PDF



Datasheets

- GW1NR-LV9 FPGA
- LT8616SXB
- SY8089
- SY8303
- LP5907

Softwares

Arduino

- Module13.2 Display Arduino Example

UiFlow1

- Module13.2 Display UiFlow1 Docs

UiFlow2

- Module13.2 Display UiFlow2 Docs

Easyloader

Easyloader	Download Link	Notes
Module13.2 Display Example Easyloader	download	/